

Stochastic Epistemic Dynamics, Attractor Geometry, and Asymptotic Convergence to a Maximal Correctness Limit Object

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Abstract

We develop a continuous stochastic dynamical systems framework for epistemic agents operating under layered constraints (L1–L3), combining Bayesian inference, social attractor dynamics, and stochastic perturbations. We analyze convergence properties toward a limit object of maximal epistemic correctness G , derive stability thresholds, and compute first-passage times under noisy epistemic conditions.

The paper ends with “The End”

1 Epistemic State Space

Let each agent A_i have belief state:

$$B_i(t) \in \mathbb{R}^3$$

corresponding to layers:

$$(L1, L2, L3)$$

Define the limit object:

$$G = (1, 1, 1)$$

Weighted epistemic distance:

$$d(B_i, G) = \sqrt{\sum_{k=1}^3 w_k (B_i^{(k)} - G^{(k)})^2}$$

with:

$$w_1 = 0.6, \quad w_2 = 0.3, \quad w_3 = 0.1$$

2 Stochastic Dynamical System

We model epistemic evolution as an SDE:

$$dB_i = (1 - \alpha)w \odot (G - B_i) dt + \alpha w \odot (S(B) - B_i) dt + \sigma dW_t^{(3)}$$

where noise acts only on L3:

$$dW_t^{(3)} = (0, 0, dW_t)$$

3 Social Attractor Field

We define:

$$S(B) = \beta_1 A_1 + \beta_2 A_2 + (1 - \beta_1 - \beta_2) \bar{B}$$

with:

$$A_1 = (1, 1, 1), \quad A_2 = (0.2, 0.8, 0.3)$$

4 Stationary Distribution of L3

L3 dynamics reduce to Ornstein–Uhlenbeck form:

$$dX_t = -w_3(X_t - \mu) dt + \sigma dW_t$$

Stationary distribution:

$$X_\infty \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{2w_3}\right)$$

5 Expected Alignment Probability

Define:

$$P(B_i, G) = e^{-\lambda d(B_i, G)}$$

Then:

$$\mathbb{E}[P] \approx e^{-\lambda \sqrt{w_3 \text{Var}(B^{(L3)}) + \sum_{k=1}^2 w_k (B^{(k)} - G^{(k)})^2}}$$

6 First-Passage Time

Let T be the time to reach G from false attractor A_2 .

For OU approximation:

$$\mathbb{E}[T] \sim \frac{1}{w_3} \int_{A_3}^1 \exp\left(\frac{w_3(y^2 - A_3^2)}{\sigma^2}\right) dy$$

Key scaling:

$$T \uparrow \text{ as } w_3 \downarrow, \quad T \downarrow \text{ as } \sigma \uparrow$$

7 Stability Threshold

Define critical social influence:

$$\alpha_c \approx \frac{w_1}{w_1 + w_3 \Delta}$$

where:

$$\Delta = |A_3 - G_3|$$

Regimes:

- $\alpha < \alpha_c$: convergence to G
- $\alpha > \alpha_c$: metastable trapping

8 Lyapunov Analysis

Define:

$$V(B) = d^2(B, G)$$

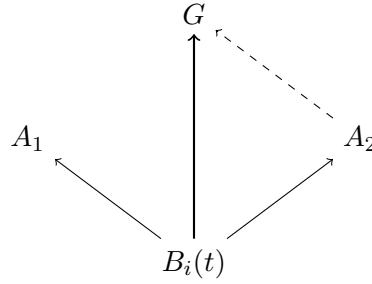
Then:

$$\frac{dV}{dt} = -2(1 - \alpha) \sum_k w_k (B^{(k)} - G^{(k)})^2 + \text{noise terms}$$

Expectation:

$$\mathbb{E}\left[\frac{dV}{dt}\right] < 0 \quad \text{if } \alpha < \alpha_c$$

9 Geometric Interpretation



10 Interpretation

- G : maximal epistemic correctness (limit object)
- L1: strong constraint field (physical truth)
- L2: emergent statistical structure
- L3: noisy narrative layer (high epistemic variance)
- α : social coupling strength
- σ : epistemic uncertainty / "vat-like" distortion

11 Conclusion

Epistemic agents evolve under a stochastic flow combining Bayesian contraction, social attractor fields, and structured noise. The system admits a globally attractive fixed point G under bounded social influence and finite L3 variance. However, metastable attractors and stochastic perturbations generate complex transient epistemic regimes.

References

- [1] T. Bayes, *An Essay towards solving a Problem in the Doctrine of Chances*, 1763.
- [2] K. Gödel, *On Formally Undecidable Propositions*, 1931.
- [3] J. Pearl, *Causality*, Cambridge University Press.
- [4] A. Kolmogorov, *Foundations of Probability Theory*, 1933.
- [5] K. Itô, *Stochastic Differential Equations*, 1944.

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